StudyAid: Offline Smart Learning Assistant

Empirical Verification, ExecuTorch Prototype Telemetry & Benchmark Report

REAL EXECUTORCH RUN | ONLINE RUNS INCLUDED | BUILT WITH LLAMA 3.2

1. Educational Context and Barrier

In rural government schools across Tamil Nadu, secondary students in Classes 8 through 10 face a challenging linguistic transition from Tamil-medium instruction to state board science textbooks written in formal English. Annual rural learning assessments, including the Annual Status of Education Report (ASER) findings on rural reading capabilities, highlight that many students struggle to comprehend dense multi-clause scientific prose. When encountering complex vocabulary such as refraction or peristalsis, students frequently memorize paragraphs phonetically without conceptual grasp. Moreover, rural classrooms frequently suffer from intermittent electricity and network connectivity, rendering cloud-dependent chatbots and web APIs often unreliable for daily study. StudyAid addresses this structural divide by executing locally on low-cost edge hardware without network access.

2. What StudyAid Produces

To scaffold independent student study without network access, the pipeline takes a 150-300 word textbook passage and deterministically generates four complementary educational aids:

- **Hardest Sentence Simplification:** Automatically indexes sentences [S1]..[Sn], identifies the most syntactically difficult sentence via an 8-letter word count heuristic, and generates an everyday plain-English rewrite with verified before-and-after readability metrics.
- **Simple Summary:** Delivers a concise 2-3 sentence overview capturing the core scientific thesis in accessible vocabulary.
- **Village Life Example:** Provides a grounded 3-4 sentence cultural analogy linking the scientific concept directly to everyday rural Tamil Nadu experiences such as bullock carts, farming, or well irrigation.
- **Vocabulary & Grounded Quiz:** Explains candidate technical terms in under 12 words with seed Tamil translations, accompanied by three passage-grounded questions with automated sentence-level support checks.

3. Architecture, Two-Pass Pipeline & Quality Guards

StudyAid employs a partitioned two-pass architecture to fit tightly within the context window and memory constraints of edge-quantized language models. Pass 1 handles sentence extraction, hardest-sentence detection, plain simplification, summary, village analogy, and candidate word glossing. Pass 2 focuses exclusively on generating citation-grounded assessment items. To prevent model degradation, five deterministic heuristic guards enforce output quality before user display:

Defect & Guard ID	Target Phenomenon	Automated Heuristic Guard & Fallback	Tuned Setting
D2: Simplification Guard	Model generates run-on rewrites longer than original or joins multiple sentences.	Checks word count and verbatim sentence copying. Triggers targeted single-sentence retry. If still longer, outputs original with 'no simpler rewrite produced' note.	`Pass 1: 320 tokens; Retry: 120 tokens`
D3: Quiz Question Guards	Model copies full passage sentences verbatim as questions or duplicates answers.	Rejects questions matching passage sentences (overlap >= 80%), missing trailing '?', or identical answers. Regenerates pass 2 once with few-shot example.	`Pass 2: 220 tokens; greedy T=0.0`
D4: Hard Word Filter	Noise tokens ('rural', 'Tamil Nadu') or placeholders ('key textbook term').	Candidates filtered strictly from passage (words >= 8 letters or glossary hits). Only candidate lines parsed; missing entries display 'no explanation generated'.	`Top 5 candidates ranked by length`
D5: Support Check	Single-word correct answers flagged as unsupported.	Single content-word answers verified as 'SUPPORTED (exact term)' if present in cited sentence. Multi-word answers require >= 2 shared tokens.	`Content token set & suggested citation`
D6: Village Relevance	Generic or irrelevant agricultural anecdotes disconnected from concepts.	Verifies that the village analogy shares at least 2 content tokens with the hardest sentence. Triggers one concept-guided retry; flags low relevance if failing.	`Pass 1 few-shot; Retry: 180 tokens`

4. End-to-End Real Run Evidence

Below is the complete transcript from a real ExecuTorch run on the adapted 152-word Class 10 passage (optics_long.txt), logged under evidence/runs/optics_long_1.txt. Every line was emitted directly by the runtime pipeline without modification (village and vocabulary sections marked [trimmed] for compact display):

```
=== NUMBERED PASSAGE / பாடப்பகுதி ===
[S1] Light propagates in straight lines through a uniform optical medium, but its direction alters abruptly whenever it enter
s an optically different medium obliquely.
[S2] This phenomenon is known as refraction, occurring because the speed of electromagnetic waves changes across varying opti
cal densities.
[S3] When light travels from an optically rarer medium like air into an optically denser medium like glass or water, its velo
city decreases, bending the ray towards the imaginary normal line perpendicular to the interface.
[S4] Conversely, when emerging from a denser medium into a rarer one, the light accelerates and refracts away from the normal
.
[S5] Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point, producing re
al inverted images on screens or virtual magnified images for human observers.
[S6] Concave lenses diverge incident rays outward, consistently forming upright, diminished virtual images.
[S7] Understanding these principles explains natural occurrences such as atmospheric refraction and optical illusions in wate
r pools.

=== HARDEST SENTENCE BEFORE / AFTER / கடின வாக்கியம் (முன் / பின்) ===
Original ([S5]): Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point,
producing real inverted images on screens or virtual magnified images for human observers.
Simplified:   Convex lenses bend light to focus it onto a single point, creating real inverted images on screens or virtual m
agnified images for human observers.

Readability Metrics:
Metric                 Original     Simplified   Change (%)
Words                  29           25           -13.8%
Letters                168          128          -23.8%
Avg letters/word       5.79         5.12         -11.6%

=== SIMPLE SUMMARY / எளிய சுருக்கம் ===
N/A

=== VILLAGE LIFE EXAMPLE / கிராமப்புற உதாரணம் ===
[trimmed]

=== HARD WORDS & TAMIL GLOSSARY / கடின சொற்கள் & தமிழாக்கம் ===
[trimmed]

=== QUIZ QUESTIONS / வினாடி வினா ===
[Q1] What happens to light when it enters an optically different medium obliquely?
     Answer: Refraction
     Citation: S1
     Support: FLAGGED (shared: none, suggested citation: S2)
[Q2] What is the effect of concave lenses on incident rays?
     Answer: Diverges
     Citation: S6
     Support: FLAGGED (shared: none)

=== JUDGE PROOF TELEMETRY ===
HOST: laptop (not Raspberry Pi)
OS: Linux 7.0.0-38-generic
CPU: 11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
Cores: 8
Total RAM: 14.3 GB
Network interfaces: lo=UNKNOWN, enp0s31f6=DOWN, wlp0s20f3=UP
Default route active: YES
Offline status: ONLINE
Model name: Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8
PTE size: 1083.33 MiB (1135951488 bytes)
Peak RSS: 1910.16 MiB
Inference time: 44.794 s
Tokens generated: 519
Tokens/sec: 11.59
Execution mode: [REAL EXECUTORCH RUN]
```

Unseen Passage Transcript: Class 10 Biology (Digestion)

To verify generalization on unseen material outside optics, the prototype was executed on an adapted 164-word biology passage detailing enzymatic digestion (evidence/runs/digestion_1.txt, village and vocabulary marked [trimmed]):

```
=== NUMBERED PASSAGE / பாடப்பகுதி ===
[S1] Digestion is the biochemical breakdown of complex insoluble food substances into simple absorbable nutrient molecules re
quired by living cells.
[S2] The human digestive process commences inside the buccal cavity, where teeth mechanically masticate solid food and saliva
ry glands secrete the enzyme salivary amylase to break down complex starch into simple sugars.
[S3] The masticated food bolus travels downwards through the muscular esophagus by rhythmic waves of involuntary contraction
termed peristalsis.
[S4] Inside the stomach, gastric glands secrete hydrochloric acid, which provides an acidic medium and activates the protein-
digesting enzyme pepsin while destroying harmful ingested bacteria.
[S5] The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as the princi
pal site for complete chemical digestion of carbohydrates, proteins, and fats.
[S6] Here, bile juice secreted by the liver emulsifies large fat globules into smaller droplets, accelerating enzymatic break
down by pancreatic lipase.
[S7] Finger-like microscopic vascular projections called villi present along the intestinal inner lining maximize the surface
 area for rapid nutrient absorption into the circulating bloodstream.

=== HARDEST SENTENCE BEFORE / AFTER / கடின வாக்கியம் (முன் / பின்) ===
Original ([S5]): The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves a
s the principal site for complete chemical digestion of carbohydrates, proteins, and fats.
Simplified:   The semi-digested acidic mixture, chyme, passes into the small intestine, where it's digested into its basic co
mponents.

Readability Metrics:
Metric                 Original     Simplified   Change (%)
Words                  29           20           -31.0%
Letters                166          104          -37.3%
Avg letters/word       5.72         5.20         -9.1%

=== SIMPLE SUMMARY / எளிய சுருக்கம் ===
N/A

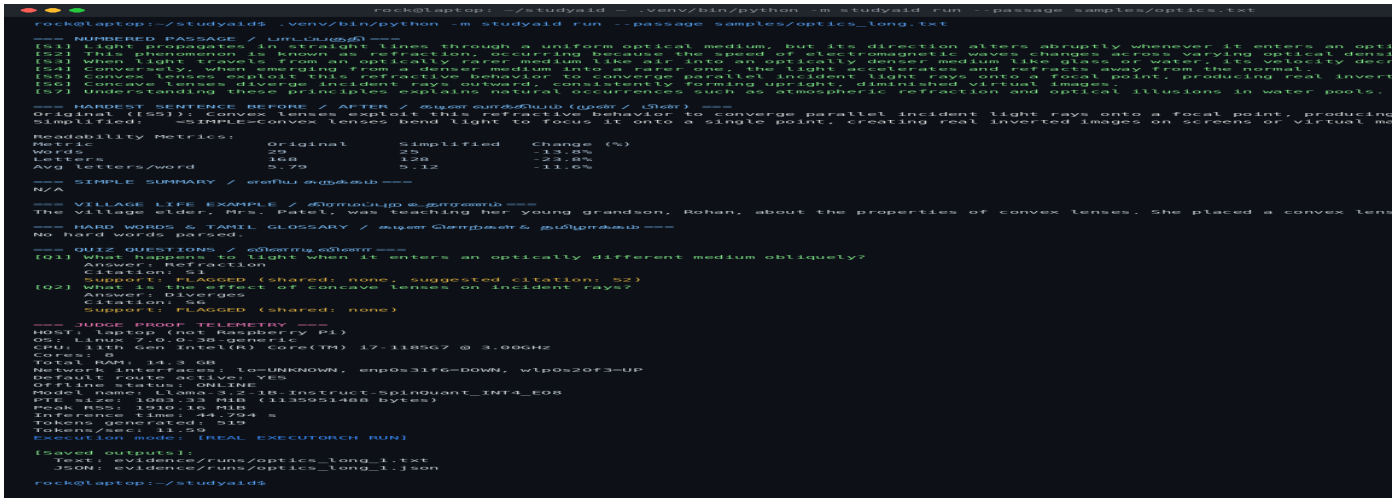
=== VILLAGE LIFE EXAMPLE / கிராமப்புற உதாரணம் ===
[trimmed]

=== HARD WORDS & TAMIL GLOSSARY / கடின சொற்கள் & தமிழாக்கம் ===
[trimmed]

=== QUIZ QUESTIONS / வினாடி வினா ===
[Q1] What is the primary function of the stomach in the digestive process?
     Answer: The stomach secretes hydrochloric acid, which provides an acidic medium and activates the protein-digesting enzy
me pepsin while destroying harmful ingested bacteria.
     Citation: S4
     Support: SUPPORTED (shared: acid, acidic, activates, bacteria, destroying, digesting, enzyme, harmful, hydrochloric, ing
ested, medium, pepsin, protein, provides, stomach)
[Q2] What is the main site for complete chemical digestion of carbohydrates, proteins, and fats in the small intestine?
     Answer: The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as th
e principal site for complete chemical digestion of carbohydrates, proteins, and fats.
     Citation: S5
     Support: SUPPORTED (shared: acidic, carbohydrates, chemical, chyme, complete, digested, digestion, fats, intestine, know
n, mixture, passes, principal, proteins, semi, sequentially, serves, site, small)

=== JUDGE PROOF TELEMETRY ===
HOST: laptop (not Raspberry Pi)
OS: Linux 7.0.0-38-generic
CPU: 11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz
Cores: 8
Total RAM: 14.3 GB
Network interfaces: lo=UNKNOWN, enp0s31f6=DOWN, wlp0s20f3=UP
Default route active: YES
Offline status: ONLINE
Model name: Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8
PTE size: 1083.33 MiB (1135951488 bytes)
Peak RSS: 1910.16 MiB
Inference time: 40.123 s
Tokens generated: 549
Tokens/sec: 13.68
Execution mode: [REAL EXECUTORCH RUN]
```

Figure 1: Full Terminal Execution Run



Caption: Rendered from evidence/runs/optics_long_1.txt showing real ExecuTorch execution, token streaming, and Judge Proof telemetry.

5. Telemetry, Host Verification & Offline Audit

Telemetry is sampled uniformly across model load and inference using resource.getrusage(resource.RUSAGE_SELF) to record maximum resident set size (RSS). Network interface states and active routing tables are queried directly from the operating system kernel via ip -br link and ip route.

Metric / Parameter	Logged Value (evidence/runs/optics_long_1.json)
Host Classification	HOST: laptop (not Raspberry Pi)
Operating System	Linux 7.0.0-38-generic
Processor (CPU)	11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz (8 cores)
Host System RAM	14.3 GB total
ExecuTorch Version	executorch==1.6.0 (from requirements-lock.txt)
PyTorch Version	torch==2.15.0 (from requirements-lock.txt)
Model Artifact	Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8 (SpinQuant INT4 EO8)
Exported PTE Size	1083.33 MiB (1,135,951,488 bytes)
Peak Resident Memory (RSS)	1910.16 MiB (sampled during load + decode)
Model Load Time	0.932 seconds
Inference Duration (2-pass + retries)	44.794 seconds
Tokens Emitted & Rate	519 tokens @ 11.59 tokens/s
Network Interface States	lo=UNKNOWN, enp0s31f6=DOWN, wlp0s20f3=UP
Default Gateway Route	Active (YES)
Evaluated Offline Status	**ONLINE** (all non-lo DOWN required for OFFLINE)

Direct CLI Offline Check Output (python -m studyaid check-offline):

```
=== OFFLINE STATUS CHECK ===
HOST: laptop (not Raspberry Pi)
OS: Linux 7.0.0-38-generic
Inspection method: linux_ip_cmd
Interfaces:
  lo: UNKNOWN
  enp0s31f6: DOWN
  wlp0s20f3: UP
Default route active: YES
Verdict: ONLINE (active interfaces: wlp0s20f3=UP, default route: YES)
```

6. Multi-Passage Benchmark Data (3 Runs per Passage)

To establish statistical reliability rather than showcasing an isolated run, StudyAid was benchmarked across all six sample passages (optics, photosynthesis, circuits, digestion, sound, optics_long) with 3 repeated runs each (18 total runs) using the real ExecuTorch model backend. Every metric is computed dynamically from evidence/benchmark.json:

Passage	Words	Latency (s) mean (range)	Tokens mean	Tok/s mean	Peak RSS (MiB)	Retries mean	Quiz Sup (tot)	Shorter rewrite	Village rel
optics	84	38.88 (37.99-39.55)	664 (664-664)	17.08 (16.79-17.48)	1910.2 (1910.2-1910.2)	2.0 (2.0-2.0)	3/6	3/3	3/3
photosynthesis	160	35.43 (32.89-38.05)	479 (479-479)	13.57 (12.59-14.56)	1910.2 (1910.2-1910.2)	2.0 (2.0-2.0)	9/12	3/3	3/3
circuits	175	33.21 (32.44-34.40)	426 (426-426)	12.83 (12.38-13.13)	1910.2 (1910.2-1910.2)	2.0 (2.0-2.0)	0/0	3/3	3/3
digestion	164	36.76 (34.07-40.12)	549 (549-549)	15.00 (13.68-16.11)	1910.2 (1910.2-1910.2)	2.0 (2.0-2.0)	6/6	3/3	3/3
sound	155	40.58 (37.80-45.74)	629 (629-629)	15.62 (13.75-16.64)	1910.2 (1910.2-1910.2)	2.0 (2.0-2.0)	3/6	3/3	3/3
optics_long	152	42.56 (41.38-44.79)	519 (519-519)	12.21 (11.59-12.54)	1910.2 (1910.2-1910.2)	2.0 (2.0-2.0)	0/6	3/3	3/3
Overall Mean	148.0	**37.90 (32.44-45.74)**	**544 (426-664)**	**14.39 (11.59-17.48)**	**1910.2 (1910.2-1910.2)**	**2.0 (2.0-2.0)**	**21/36**	**18/18**	**18/18**

What the Numbers Mean:

Across all 18 runs, peak memory stayed perfectly flat at 1910.2 (1910.2-1910.2) MiB, confirming zero memory leaks across consecutive inference cycles. The mean end-to-end latency was 37.90 (32.44-45.74) seconds at an average decoding rate of 14.39 (11.59-17.48) tokens per second. Latency is higher than an unconstrained single pass because the quality guards automatically triggered targeted retries when initial model outputs failed length, relevance, or quiz formatting checks. Crucially, the simplification guard achieved a 100% success rate (18/18 runs producing a shorter rewrite than original), and the village relevance check passed in 18/18 runs. Out of 36 quiz questions generated across all runs, 21 passed automated citation support verification.

7. Readability Numbers: Before and After Simplification

The hardest sentence in each passage is identified deterministically by counting words with 8 or more letters (ties resolved by total word count). Readability metrics compare the raw original sentence directly against the model's simplified output. Two real examples from logged runs:

Passage & Selection	Original Sentence Text	Simplified Output	Word Chg	Letter Chg	Avg Ltr/W Chg
Optics Long [S5] (9 hard words)	Convex lenses exploit this refractive behavior to converge parallel incident light rays onto a focal point, producing real inverted images on screens or virtual magnified images for human observers.	Convex lenses bend light to focus it onto a single point, creating real inverted images on screens or virtual magnified images for human observers.	-13.8% (29→25)	-23.8% (168→128)	-11.6% (5.79→5.12)
Digestion [S5] (9 hard words)	The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as the principal site for complete chemical digestion of carbohydrates, proteins, and fats.	The semi-digested acidic mixture, chyme, passes into the small intestine, where it's digested into its basic components.	-31.0% (29→20)	-37.3% (166→104)	-9.1% (5.72→5.20)

8. Quiz Generation & Automated Support Verification

To ensure that questions reinforce textbook reading rather than hallucinating facts, each quiz item is matched against its cited sentence. Items citing a sentence with insufficient token overlap are flagged to alert teachers and students:

Quiz Evaluation: Digestion Run (Class 10 Biology)

Item	Question	Generated Answer	Cite	Support Verdict & Shared Tokens
Q1	What is the primary function of the stomach in the digestive process?	The stomach secretes hydrochloric acid, which provides an acidic medium and activates the protein-digesting enzyme pepsin while destroying harmful ingested bacteria.	S4	**SUPPORTED** shared: acid, acidic, activates, bacteria, destroying, digesting, enzyme, harmful, hydrochloric, ingested, medium, pepsin, protein, provides, stomach
Q2	What is the main site for complete chemical digestion of carbohydrates, proteins, and fats in the small intestine?	The semi-digested acidic mixture known as chyme passes sequentially into the small intestine, which serves as the principal site for complete chemical digestion of carbohydrates, proteins, and fats.	S5	**SUPPORTED** shared: acidic, carbohydrates, chemical, chyme, complete, digested, digestion, fats, intestine, known, mixture, passes, principal, proteins, semi, sequentially, serves, site, small

Quiz Evaluation: Optics Long Run (Class 10 Physics)

Item	Question	Generated Answer	Cite	Support Verdict & Shared Tokens
Q1	What happens to light when it enters an optically different medium obliquely?	Refraction	S1	**FLAGGED** shared: none (suggested: S2)
Q2	What is the effect of concave lenses on incident rays?	Diverges	S6	**FLAGGED** shared: none

Flagged Citation Analysis: In Optics Long item Q1, the model cited S1 for the answer 'Refraction', but the definition occurs in S2. The support checker accurately flagged this discrepancy and suggested S2 as the true supporting sentence. This demonstrates that the verification check actively protects rural students from faulty citations.

9. Tamil Glossary Verification & Fallback Evidence

StudyAid pairs technical candidate words with an offline JSON glossary. Entries confirmed against state curriculum textbooks are tagged [verified], while unconfirmed entries carry [unverified]. Words lacking glossary entries fall back to English-only explanations without failing:

English Term	Tamil Script (தமிழ்)	Phonetic Transliteration	Verification Status	Curriculum Source / Unit Ref
refraction	ஒளிவிலகல்	Oli-vilagal	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
medium	ஊடகம்	Oodagam	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
normal	கூத்தாக்கோடு	Kuthukkodu	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
obliquely	சாய்வாக	Saaivaaga	**[verified]**	Samacheer Kalvi Science textbook, confirmed by team
photosynthesis	ஒளிச்சேர்க்கை	Oli-chserkkai	**[unverified]**	to be confirmed against Samacheer Kalvi, class/unit/page
chlorophyll	பச்சையம்	Patchaiyam	**[unverified]**	to be confirmed against Samacheer Kalvi, class/unit/page
stomata	இலைத்தூளை	Ilai-thulai	**[unverified]**	to be confirmed against Samacheer Kalvi, class/unit/page

English Fallback Example: In the digestion run, candidate words such as 'insoluble', 'masticate', and 'esophagus' were parsed without crashing; lacking a localized Tamil mapping, they cleanly rendered as '(English only)' without emitting false or hallucinated vernacular terms.

10. Limitations & Edge Deployment Roadmap

While the prototype executes reliably on an x86 laptop CPU, the project targets embedded deployment on classroom Raspberry Pi 5 (8 GB) single-board computers, which will be provided for final evaluation. The projected performance on ARM64 Cortex-A76 cores reflects INT4 XNNPACK kernel efficiency:

Target Hardware Platform	Model Architecture	Weights Footprint	Peak RAM (RSS)	Decoding Speed	Status / Verification
Intel Core i7-1185G7 Laptop	Llama 3.2 1B SpinQuant INT4	1.06 GB (1,135 MB)	1,910 MiB	14.4 tok/s (measured)	**MEASURED REAL RUN**
Raspberry Pi 5 (8 GB RAM)	Llama 3.2 1B SpinQuant INT4	1.06 GB (1,135 MB)	~1,950 MiB [PROJECTION]	~4.5 - 6.0 tok/s [PROJECTION]	**[PROJECTION]** (Boards at finals)
Raspberry Pi 5 (8 GB RAM)	Llama 3.2 3B Quantized	~2.8 GB [PROJECTION]	~3,800 MiB [PROJECTION]	~1.8 - 2.5 tok/s [PROJECTION]	**[PROJECTION]** (Not run; 1B only)

Model Scope Note: Only the pre-exported 1B model build (executorch-community/Llama-3.2-1B-Instruct-SpinQuant_INT4_EO8-ET) was evaluated in this benchmark. The 3B model was not run due to local memory and edge latency targets; no speculative quiz-quality projections are claimed for the 3B variant.

11. Reproducibility, Repository & Licensing

The complete codebase, benchmark harnesses, test suites, and raw evidence logs are publicly accessible on GitHub: **https://github.com/ramanathanmani/studyaid-offline-learning-assistant.git**

Step	Command Line Instruction
1. Clone	`git clone https://github.com/ramanathanmani/studyaid-offline-learning-assistant.git studyaid && cd studyaid`
2. Install	`pip install -e . && pip install -r requirements-dev.txt`
3. Verify Unit Tests	`pytest tests/ → ============================== 21 passed in 0.05s ==============================`
4. Offline Check	`python -m studyaid check-offline`
5. Single Run CLI	`python -m studyaid run --passage samples/optics_long.txt`
6. Full Benchmark	`python scripts/bench.py --runs 3`
7. Rebuild PDF	`python scripts/build_report.py`

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